CS & IT ENGINEERING

THEORY OF COMPUTATION

Grammar

[redacted]

Lecture - 1

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Recap of Previous Lecture

Topic ?????

Finite Automata

Regular Expression

Regular languages

Closure properties of Regular language

Topics to be Covered

Topic — Grammar

Topic ?? Context free Grammar

Topic ?? Regular Grammar

Topic ?? Context free Grammar

Topic : Grammar

C, C++, java

- Set of rules used to describe strings of a language is known as grammar.
- Formal definition of grammar is

$\{\boxed{G=(N, T, P, S)}\}$

- N :- non terminals or variables $= \{S, A, B, C\} = 4$
- T :- Terminals $= \{a, b, d\} = 3$
- P :- no. of productions $= 4$
- S :- Starting symbol

Ex:-

1) $S \rightarrow ABC$
2) $A \rightarrow a \mid b$ } 2
3) $B \rightarrow b \mid c$ } 2
4) $C \rightarrow d \mid a$ } 2

$A \rightarrow a$
$A \rightarrow b$

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$$4\ \text{product}\left\{\begin{array}{l} \overbrace{S \to I = E}^{1} \\ \\ E \to \underbrace{E+E}_{2} \mid \underbrace{E*E}_{3} \mid \underbrace{a}_{4} \\ I \to a \end{array}\right.$$

No. of productions = ?

Non terminals : 3 $\{S, I, E\}$

Topic : Grammar

- For every language grammar exit & every grammar generates one language.
- All grammars is of a from α replacement of $\boxed{\alpha \rightarrow \beta}$, where β is replacement of α

$$\boxed{\alpha \longrightarrow \beta}$$

$L_1 = \{a^n b^n \mid n \geq 1\}$ ✓

$L_2 = \{a^n b^n c^n \mid n \geq 1\}$ ✓

Topic : Derivation

- The process of deriving strings from the given grammar known as derivation.
- The derivation can be either left most derivation or right most derivation

- **Left most derivation:**
  It is the derivation in which left most non terminal is replaced by its R.H.S part at every step.

- **Right most derivation:**
  It is a derivation in which right most non terminal is replaced by its R.H.S part at every step.

Derivation Tree (or) Parse Tree

- Tree representation of the derivation is known as derivation tree.
- All leaf node of the parse tree is known as yield of parse tree .
- while reading yield from left to right sentence of the grammar can be generate.

Sentential form

- Each step in the derivation is one sentential form.
- Hence sentential form is combination of terminals & non terminals (sentence also can be included)
- If the derivation is left most then sentential form is left sentential form.
- If the derivation is right most then sentential is right sentential form
- Every grammar represents only one language but for one language more than one grammar may exist.
- For regular languages there exist a grammar known as regular grammar.

- Context free language there exist a grammar known as context free grammar.
- Context sensitive language there exist a grammar known as context sensitive grammar.
- For recursive enumerable language there exist a grammar known as unrestricted grammar.

Type	Language (Grammers)	From of Productions	Accepting Dived
3	Regular	$A \rightarrow aB$, $A \rightarrow A$	Finite Automaton
2	Context –free	$A \rightarrow \alpha$	Pushdown Automaton
1	Context sensitive	$A \rightarrow \beta$ With $\lvert\alpha\rvert \geq \lvert\beta\rvert$	LBA
0	Unrestricted	$\alpha \rightarrow \beta$	Turing machine

THANK - YOU